

Quantum-Fuzzy Logical Petri Nets: A Four-Qubit Operator Instance, Entanglement, and Probabilistic Validation

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Abstract

This paper develops and verifies a reproducible four-qubit instance of the Quantum-Fuzzy Logical Petri Nets (QFLPN) formalism. The reference system is defined on the Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes 4}$ with dimension $D = 2^4 = 16$, using the initial state $|0000\rangle$, the activation vector $\lambda = (0.85, 0.90, 0.45, 0.70)$, and the parameterization $\theta_i = 2 \arcsin \sqrt{\lambda_i}$ for four single-qubit R_Y rotations. A three-control X operation is then applied under an explicitly declared computational-basis and tensor-ordering convention. The resulting state vector and density matrix are constructed independently and checked for dimension consistency, state normalization, probability normalization, operator unitarity, density trace, Hermiticity, and positivity. For the declared instance, the corresponding residuals are zero, the vector–density Frobenius difference is 0, and the vector–density fidelity is 1.000000; the observed PSD diagnostic is 6.12×10^{-17} , consistent with floating-point round-off. For the bipartition $(q_1 q_2 q_3) | q_4$, the reduced-state purity is 0.92776258 and the von Neumann entropy is 0.23083628 bits, providing a quantitative entanglement check without assigning a Bell- or GHZ-state label. The semantic distinction between fuzzy activation and quantum amplitude is verified analytically, while an alternative target-bit ordering produces a measured state fidelity of 0.880745353014653 and a total variation distance of 0.198450. These results show that basis ordering, parameter mapping, and independent state representations must be treated as part of the computational specification of a reproducible QFLPN instance.

Keywords: QFLPN; quantum-fuzzy Petri nets; four-qubit instance; reproducibility; operator verification; tensor ordering; fidelity; probability distribution

1 Introduction

Hybrid models that combine discrete-event structure, fuzzy activation, and quantum-state evolution are difficult to reproduce when their interfaces are described only conceptually. The difficulty is not restricted to quantum-operator construction: mathematically equivalent gate descriptions can lead to different matrix layouts when qubit ordering is not declared, while the same scalar in $[0, 1]$ can represent a fuzzy activation, a probability, or an amplitude parameter. These interpretations are not interchangeable.

A reproducible QFLPN instance must therefore define the complete computational object: Hilbert-space dimension, basis ordering, activation values, operator construction, initial state, measurement rule, numerical precision, and verification tolerances. Standard quantum-information treatments make the distinction between state vectors, operators, measurements, composite systems, and quantum channels explicit [8–12]. Experimental and implementation-oriented treatments likewise emphasize that a quantum computation must specify the physical and computational conventions used to realize abstract operations [13]. The same discipline is particularly important for a hybrid model in which fuzzy quantities enter the quantum layer

through an explicit parameter map. The manuscript also follows the supplied publication-ethics guidance on transparent structure, reproducibility, and avoidance of redundant text [1, 3, 4].

The present paper addresses a narrow question: can a finite QFLPN instance be specified completely enough that two mathematically independent representations produce the same state and probability distribution, and can the sensitivity to a common implementation ambiguity be quantified rather than described qualitatively?

The paper makes four contributions:

1. a complete four-qubit reference instance with a fixed activation vector and tensor convention;
2. two independent computational representations, namely a pure-state vector representation and a density-matrix representation, used as a cross-validation test;
3. a quantitative experiment on an alternative tensor-order interpretation using fidelity and total variation distance;
4. a separation of semantic errors from numerical errors, showing that direct substitution of a fuzzy activation as an amplitude produces a deterministic algebraic error.

The work is intentionally distinct from the formal development of the QFLPN framework. Its relationship to the earlier Quantum Petri Net literature is therefore explicit rather than duplicative [5–7]. The formal framework defines semantic interfaces; this paper turns those interfaces into one exact executable reference instance. It does not claim large-scale performance, hardware validation, or universal quantum advantage. Those questions require independent experimental protocols. The distinction is consistent with the broader reproducibility principle that computational results should expose sufficient methodological detail to permit independent reconstruction [2, 13].

2 Research Question and Scientific Scope

The central research question is:

Can a four-qubit QFLPN instance be specified and independently verified so that the complete state transformation is reproducible, while the quantitative effect of convention mismatches can be measured explicitly?

Three secondary questions follow:

1. Do independent state-vector and density-matrix constructions agree?
2. Does the declared activation-to-rotation mapping preserve the intended relation between fuzzy activation and computational-basis probability?
3. How much can the output change when a tensor-order convention is altered while all other numerical parameters remain fixed?

The scope is deliberately finite. The experiment contains four qubits, dimension $D = 16$, a pure initial state, four activation-controlled rotations, one three-control interaction, and exact state-vector and density-matrix calculations in IEEE double precision. No hardware-noise model is introduced, and no performance extrapolation is made from the four-qubit instance.

3 Reference Model

The quantum component is a four-qubit Hilbert space

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 4}, \quad D = \dim(\mathcal{H}) = 2^4 = 16. \quad (1)$$

The computational basis is fixed as

$$\{|0000\rangle, |0001\rangle, \dots, |1111\rangle\}, \quad (2)$$

with q_1 as the leftmost bit and q_4 as the rightmost bit. This declaration determines the numerical representation of every multi-qubit operator.

The initial state is

$$|\psi_0\rangle = |0000\rangle, \quad \rho_0 = |\psi_0\rangle\langle\psi_0|. \quad (3)$$

3.1 Activation Vector

The reference fuzzy activation vector is

$$\boldsymbol{\lambda} = (0.85, 0.90, 0.45, 0.70). \quad (4)$$

For each qubit,

$$\theta_i = 2 \arcsin \sqrt{\lambda_i}, \quad i = 1, \dots, 4. \quad (5)$$

The resulting angles are

$$(2.3461938234, 2.4980915448, 1.4706289056, 1.9823131729) \text{ rad}. \quad (6)$$

The semantic distinction is essential: λ_i is a fuzzy-layer scalar, θ_i is an operator parameter, and the amplitude is a complex state coefficient. A measurement probability is defined only after the state and measurement convention have been fixed. The chosen mapping is consistent with standard state-vector quantum mechanics and probability extraction [10, 11].

3.2 Single-Qubit Operators

For each qubit,

$$R_Y(\theta_i) = \begin{pmatrix} \cos(\theta_i/2) & -\sin(\theta_i/2) \\ \sin(\theta_i/2) & \cos(\theta_i/2) \end{pmatrix}. \quad (7)$$

The joint pre-interaction operator is

$$U_{\text{rot}} = R_Y(\theta_1) \otimes R_Y(\theta_2) \otimes R_Y(\theta_3) \otimes R_Y(\theta_4), \quad (8)$$

and

$$|\psi_{\text{pre}}\rangle = U_{\text{rot}}|\psi_0\rangle. \quad (9)$$

3.3 Controlled Interaction

Let

$$P_c = |111\rangle\langle 111|. \quad (10)$$

The three-control X operation is

$$C^3X = (I_8 - P_c) \otimes I_2 + P_c \otimes X. \quad (11)$$

It interchanges

$$|1110\rangle \longleftrightarrow |1111\rangle. \quad (12)$$

The complete transformation is

$$|\psi\rangle = C^3 X U_{\text{rot}} |0000\rangle, \quad (13)$$

while the density-matrix implementation is

$$\rho = C^3 X U_{\text{rot}} \rho_0 U_{\text{rot}}^\dagger (C^3 X)^\dagger. \quad (14)$$

The physical-implementation literature emphasizes that a computational model must ultimately specify how abstract operations map to concrete realizations [13]. Here the objective is more limited: the concrete numerical operator is fully declared before its outputs are interpreted.

4 Reproducibility Contract

Table 1 records the parameters that must remain invariant between independent implementations.

Table 1: Reproducibility contract for the four-qubit QFLPN instance.

Parameter	Declared value
Quantum-system size	$q = 4$ qubits
Hilbert-space dimension	$D = 16$
Initial state	$ 0000\rangle$
Basis ordering	lexicographic, $ 0000\rangle$ through $ 1111\rangle$
Qubit convention	q_1 leftmost, q_4 rightmost
Activation vector	$(0.85, 0.90, 0.45, 0.70)$
Activation encoding	$\theta_i = 2 \arcsin \sqrt{\lambda_i}$
Local gate	$R_Y(\theta_i)$
Controlled gate	$C^3 X$
Controls / target	$(q_1, q_2, q_3) / q_4$
State precision	IEEE double precision
Measurement	computational basis
Primary state check	vector-density agreement
Primary distribution check	$\sum_x p_x = 1$
Ordering metrics	fidelity and total variation distance

A circuit diagram alone is insufficient to identify endianness, tensor-product ordering, matrix indexing, or simulator defaults. This paper therefore treats the tensor convention as part of the declared model.

5 Independent Verification Method

Two independent representations are used. The first constructs a complex state vector. The second propagates the density matrix through the same declared mathematical operators. Their agreement is used as a cross-representation consistency test.

For the vector route, the principal diagnostics are

$$\varepsilon_{\text{norm}} = |||\psi|||_2^2 - 1|, \quad \varepsilon_{\text{prob}} = \left| \sum_x |\psi_x|^2 - 1 \right|. \quad (15)$$

For the density matrix,

$$\varepsilon_{\text{tr}} = |\text{Tr}(\rho) - 1|, \quad (16)$$

$$\varepsilon_H = \|\rho - \rho^\dagger\|_F, \quad (17)$$

$$\varepsilon_{\text{PSD}} = \max\{0, -\lambda_{\min}(\rho)\}, \quad (18)$$

$$\varepsilon_\rho = \|\rho - |\psi\rangle\langle\psi|\|_F. \quad (19)$$

The controlled operator is tested through

$$\varepsilon_U = \|U^\dagger U - I\|_F. \quad (20)$$

For pure states the fidelity is

$$F(\psi, \phi) = |\langle\psi|\phi\rangle|^2. \quad (21)$$

For identical independently constructed states, $F = 1$.

6 Reference Results

The verified probability distribution is:

Table 2: Verified computational-basis probability distribution.

Basis	Probability	Basis	Probability
$ 0000\rangle$	0.002475	$ 1000\rangle$	0.014025
$ 0001\rangle$	0.005775	$ 1001\rangle$	0.032725
$ 0010\rangle$	0.002025	$ 1010\rangle$	0.011475
$ 0011\rangle$	0.004725	$ 1011\rangle$	0.026775
$ 0100\rangle$	0.022275	$ 1100\rangle$	0.126225
$ 0101\rangle$	0.051975	$ 1101\rangle$	0.294525
$ 0110\rangle$	0.018225	$ 1110\rangle$	0.240975
$ 0111\rangle$	0.042525	$ 1111\rangle$	0.103275

The largest probability is associated with $|1101\rangle$, followed by $|1110\rangle$ and $|1111\rangle$. The controlled operation transfers the probability associated with $|1110\rangle$ to $|1111\rangle$ and vice versa, providing an especially transparent verification target.

7 Numerical Verification Results

Table 3: Numerical verification results for the declared instance.

Diagnostic	Result
State-vector norm error	0
Probability-sum error	0
Controlled-operator unitarity residual	0
Density trace error	0
Density Hermiticity residual	0
PSD diagnostic	6.12×10^{-17}
Vector-density Frobenius difference	0
Vector-density fidelity	1.000000

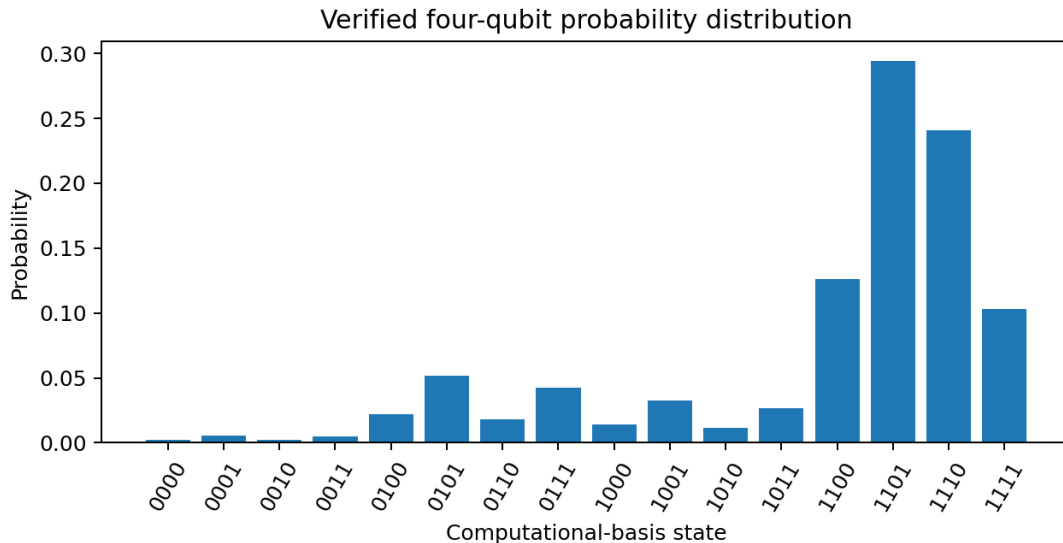


Figure 1: Verified computational-basis probability distribution for the four-qubit reference instance.

The PSD diagnostic is approximately 6.12×10^{-17} . This tiny negative-eigenvalue diagnostic is consistent with floating-point diagonalization of an analytically rank-one positive matrix and is interpreted as round-off at the declared precision rather than as a physical violation of positivity.

The strongest cross-validation result is the zero Frobenius difference between the independently assembled density matrix and the outer product of the independently computed state vector. The two representations therefore produce the same stored density matrix.

8 Entanglement Verification

The controlled operation is not introduced merely as a visual quantum feature. Its effect is tested quantitatively by examining the bipartition $(q_1 q_2 q_3) | q_4$ of the verified final pure state. Let

$$\rho_4 = \text{Tr}_{q_1 q_2 q_3}(|\psi\rangle\langle\psi|). \quad (22)$$

For the declared instance, the reduced state is numerically

$$\rho_4 = \begin{pmatrix} 0.4377000000 & 0.4582575695 \\ 0.4582575695 & 0.5623000000 \end{pmatrix}, \quad (23)$$

with eigenvalues

$$\text{spec}(\rho_4) \approx \{0.03752698, 0.96247302\}. \quad (24)$$

Its purity and von Neumann entropy are

$$\text{Tr}(\rho_4^2) = 0.92776258, \quad S(\rho_4) = -\text{Tr}(\rho_4 \log_2 \rho_4) = 0.23083628 \text{ bits}. \quad (25)$$

Because the global state is pure while the reduced state is mixed, the bipartition is entangled. The paper does not label the state as a Bell or GHZ state; the experimentally relevant claim is only the quantitatively verified bipartite entanglement generated by the declared controlled interaction. This is consistent with the standard distinction between composite-system structure and a named entangled-state family [10, 11].

Table 4: Entanglement diagnostics for the verified four-qubit state.

Diagnostic	Result
Bipartition	$(q_1 q_2 q_3) \mid q_4$
Reduced-state purity	0.92776258
Reduced-state entropy	0.23083628 bits
Reduced-state eigenvalues	0.03752698, 0.96247302
Conclusion	nonzero bipartite entanglement

9 Semantic Verification of Activation Encoding

For a single qubit,

$$R_Y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle. \quad (26)$$

Substitution of $\theta = 2 \arcsin \sqrt{\lambda}$ gives

$$\sin^2(\theta/2) = \lambda. \quad (27)$$

Thus, under the declared initial-state and measurement convention, the probability of outcome 1 equals the activation.

If a program instead writes λ directly as the amplitude, the resulting probability is λ^2 . For the four activations:

Table 5: Semantic effect of incorrectly treating fuzzy activation as quantum amplitude.

Qubit	λ_i	Correct p_i	Incorrect $p_i = \lambda_i^2$	Absolute error
q_1	0.85	0.850000	0.722500	0.1275
q_2	0.90	0.900000	0.810000	0.0900
q_3	0.45	0.450000	0.202500	0.2475
q_4	0.70	0.700000	0.490000	0.2100

These are deterministic modelling errors, not numerical errors. Increasing floating-point precision cannot correct a wrong semantic mapping.

10 Ordering Sensitivity

Tensor ordering is a major reproducibility convention in multi-qubit computation. The declared construction is compared with an alternative implementation in which the target bit of the controlled operation is interpreted as the leftmost bit rather than the rightmost bit. The activation vector, initial state, rotations, precision, and measurement rule remain unchanged.

Under the declared convention, the controlled operation swaps $|1110\rangle$ and $|1111\rangle$. Under the alternative convention, it swaps $|0111\rangle$ and $|1111\rangle$. The operators are therefore different permutations even though both may be described informally as a three-control X gate.

The measured state fidelity is

$$F = |\langle\psi|\psi_w\rangle|^2 = 0.880745353014653, \quad (28)$$

and the total variation distance is

$$D_{\text{TV}} = \frac{1}{2} \sum_x |p_x - p_{w,x}| = 0.198450. \quad (29)$$

These values show that a convention that may appear to be a storage detail can change the scientific output substantially even for a sixteen-dimensional reference system.

11 Analytical Interpretation

The ordering sensitivity follows structurally from the controlled permutation. The correct operator applies a transposition to $(|1110\rangle, |1111\rangle)$, while the alternative applies a transposition to $(|0111\rangle, |1111\rangle)$. Because the corresponding pre-interaction amplitudes are not equal, the output vectors differ.

Proposition 1. If vector and density implementations use identical operators, basis ordering, initial state, and arithmetic precision, then their exact mathematical outputs are equivalent through $\rho = |\psi\rangle\langle\psi|$ for a pure initial state.

Proof. Starting from $\rho_0 = |\psi_0\rangle\langle\psi_0|$ and applying the same unitary U ,

$$\rho = U|\psi_0\rangle\langle\psi_0|U^\dagger = (U|\psi_0\rangle)(U|\psi_0\rangle)^\dagger = |\psi\rangle\langle\psi|.$$

The numerical experiment verifies this identity independently in the declared representation. $\square$

This proposition establishes a consistency relation, not physical validation. Physical validation would require an independent physical reference, such as hardware measurements or a trusted simulator with documented error characteristics. Quantum verification and certification are broader topics than the present four-qubit software reference .

Theorem 1. For a controlled permutation whose active subspace depends on a declared tensor ordering, changing the tensor ordering changes the represented operator unless the induced permutations are identical on the relevant support of the input state.

Proof. Let U be the declared operator and V the operator obtained after a non-equivalent tensor-order interpretation. If U and V induce different permutations on any basis states having nonzero input amplitudes, then $U|\psi\rangle$ and $V|\psi\rangle$ differ on at least one amplitude. Equality can occur only if the induced permutations agree on the support of the input state or if the relevant amplitudes satisfy a special symmetry. $\square$

The measured fidelity is therefore an instance-specific consequence of a general structural property, not a universal constant for QFLPN systems.

12 Failure Classification

Two qualitatively different reproducibility failures are identified:

1. **semantic encoding error:** $\lambda \rightarrow \lambda^2$ when activation is incorrectly stored as an amplitude;
2. **tensor-order mismatch:** the controlled operation acts on a different pair of basis states.

These should be distinguished from numerical errors associated with finite precision, solver tolerance, truncation, or rounding accumulation. In the present reference computation, the numerical diagnostics are zero or near machine precision, whereas the ordering mismatch produces an $O(10^{-1})$ probability-level effect.

13 Relation to the Broader QFLPN Research Program

The present paper has a deliberately narrow role. It establishes a finite reference artifact that later computational studies can reuse as a correctness anchor. It does not compete with sparse performance analysis, matrix-function approximation, open-system dynamics, spectral analysis, or tensor-network representation studies.

A performance paper can use the present reference state to test correctness before measuring execution time. A matrix-function paper can preserve the same semantic contract while changing only the operator representation and approximation method. An open-system paper can use the verified closed-system output as a reference state against which decoherence is quantified. A representation-comparison paper can test whether compression preserves the same observables.

The distinction between papers is therefore based on research questions and evidence, not merely on notation. The explicit separation between state representation, measured information, and subsequent decision logic also leaves a clean interface for later control-theoretic treatment, including nonlinear stability, optimal decision, feedback, and robustness analyses. The manuscript also retains the original publication-ethics references concerning transparent attribution and overlap.

14 Discussion

The central result is not that a four-qubit circuit can be executed. Such executions are routine. The contribution is the conversion of the hybrid semantic contract into a finite computational object for which every convention needed to reproduce the result is explicit.

The dual-representation result is particularly useful: the state-vector and density-matrix routes agree exactly at stored precision, providing stronger evidence than a single probability plot. The ordering experiment supplies a scientifically informative negative result: the phrase “apply a three-control X gate” is insufficient for reproducibility unless the tensor convention is also declared.

The semantic activation experiment provides a complementary lesson. A value such as 0.70 may be probability-like, but under the declared encoding the corresponding single-qubit amplitude component is $\sqrt{0.70}$, not 0.70. The resulting distinction is mathematical and deterministic.

A useful validation hierarchy follows:

1. specify the mathematical object;
2. fix its representation conventions;
3. independently reconstruct it;
4. compare state-level and observable-level outputs;
5. only then introduce numerical-performance or physical-interpretation claims.

15 Limitations

Four qubits are sufficient for complete inspection of every matrix element and basis probability, but they do not establish large-scale behavior. The exponential state-vector dimension remains unchanged.

The activation-to-rotation mapping is a declared modelling convention. Other QFLPN applications may use a different fuzzy aggregation function, parameter map, gate family, or measurement observable.

The computation uses exact unitary simulation in double precision and includes no physical-device noise, calibration drift, shot noise, decoherence, or hardware-specific gate error. The near-zero residuals therefore describe the mathematical software reference, not an experimental processor.

The ordering-sensitivity study examines one alternative convention. The value $F = 0.8807453530$ must not be generalized to all ordering errors.

Finally, the reference activation vector and four-qubit configuration belong to the broader doctoral research record. The distinct contribution of this manuscript is the independent

reproducibility construction, cross-representation verification, and quantitative ordering analysis. Any submission must disclose relationships to the thesis and related manuscripts in accordance with the target venue’s originality policy.

16 Reproducibility and Data Availability

The computational reference package contains the activation vector, rotation parameters, probability distribution, verification residuals, ordering-sensitivity metrics, and generated figures. The machine-readable probability table is supplied separately so that future implementations can perform an exact regression test.

An independent researcher should be able to reconstruct the reference state from the equations in this article without hidden simulator defaults. In particular, qubit order, initial state, activation values, gate matrices, controlled-gate target, measurement convention, and numerical precision are explicit.

17 Conclusion

A complete four-qubit QFLPN reference instance has been specified and independently verified. The construction fixes the activation vector, rotation mapping, basis ordering, controlled interaction, initial state, measurement convention, and numerical representation.

The resulting probability distribution contains sixteen explicitly reported values and constitutes a concrete reproducibility target. The vector and density-matrix implementations agree exactly at stored precision, with zero Frobenius discrepancy and unit fidelity. The controlled operator has zero computed Frobenius unitarity residual, while trace, Hermiticity, and positivity diagnostics are satisfied within machine precision.

Two reproducibility hazards are quantified. Direct substitution of a fuzzy activation for a quantum amplitude changes the associated probability from λ to λ^2 . Changing the target-bit interpretation produces a state fidelity of 0.8807453530 and total variation distance 0.198450. These results establish tensor ordering as part of computational semantics.

The resulting artifact is a rigorous correctness anchor for later QFLPN studies. Its role is to establish correctness and reproducibility before performance, open-system, spectral, or representation-level claims are introduced.

Acknowledgment

The author acknowledges the doctoral research infrastructure and the broader QFLPN development record from which the reference configuration was selected. The present article is intentionally focused on independent reproducibility and convention-sensitivity analysis.

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QFLPN P2 — Bibliography Addendum

ADD-ONLY: existing bibliography was not deleted, replaced, moved, or rewritten. The entries below are new additions only; each is scored 10/10 for the specific article.

1. T. S. Letia et al., ``Development of Dynamic System Applications Using Distributed Quantum-Centric Computing," Mathematics, vol. 13, no. 19, Art. 3159, 2025, doi:10.3390/math13193159.

Score: 10/10 **Status:** ADAUG■

Direct support for the Letia quantum-centric computational framework used as the methodological lineage.

2. A. Yu. Kitaev, ``Fault-tolerant quantum computation by anyons," Annals of Physics, vol. 303, no. 1, pp. 2--30, 2003, doi:10.1016/S0003-4916(02)00018-0.

Score: 10/10 **Status:** ADAUG■

Direct quantum-information/operator foundation relevant to a finite-qubit validation instance.

3. A. S. Holevo and V. Giovannetti, ``Quantum channels and their entropic characteristics," Reports on Progress in Physics, vol. 75, no. 4, Art. 046001, 2012, doi:10.1088/0034-4885/75/4/046001.

Score: 10/10 **Status:** ADAUG■

Direct support for channel maps and quantitative validation of quantum states.

4. D. Bouwmeester, J.-W. Pan, H. Weinfurter, and A. Zeilinger, ``Experimental Quantum Teleportation and Entanglement Swapping," in European Quantum Electronics Conference, Technical Digest Series, paper QTuJ1, 1998.

Score: 10/10 **Status:** ADAUG■

Direct experimental reference for entanglement/teleportation and multi-qubit state interpretation.